



Employee Shift & Attendance Management System

Java Programming – Case Study 85

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1. Introduction

This project is a desktop application built in Java that helps a company running multiple shifts to manage its staff. It lets an administrator register employees, assign them to Morning, Evening or Night shifts, record daily check-in and check-out times, calculate worked hours, and flag irregular attendance. The user interface is built with Java Swing, and the whole program is contained in a single file, `EmployeeShiftGUI.java`.

2. Problem Statement

A company operating in multiple shifts needs a system to schedule employee shifts, record daily attendance via check-in/check-out, and calculate worked hours

across morning, evening and night shifts. The system should flag irregular attendance patterns.

3. Objectives

- Maintain employee and shift information.
- Schedule employees across shifts.
- Record daily check-in/check-out attendance.
- Calculate worked hours per shift.
- Search attendance records.
- Provide a shift management GUI.

4. Modules

No.	Module	What it does	Method(s) in code
1	Employee Registration	Add, view, update and delete employees	<code>addEmployee,</code> <code>viewEmployees,</code> <code>updateEmployee,</code> <code>deleteEmployee</code>
2	Shift Scheduling	Assign Morning/Evening/Night shift to an employee; view shift timings	<code>assignShift,</code> <code>viewShifts</code>
3	Attendance Recording	Record check-in and check-out time	<code>checkIn,</code> <code>checkOut</code>
4	Worked Hours Calculation	Calculate hours worked per attendance record	<code>calculateHours,</code> <code>Attendance.workedHours</code>
5	Irregularity Detection	Flag late arrivals and short working days	<code>irregularities</code>
6	Attendance Search	Search records by employee ID or by shift	<code>searchByEmployee,</code> <code>searchByShift</code>

No.	Module	What it does	Method(s) in code
7	Shift Reports	Per-employee summary of shift and total hours	<code>generateReport</code>

5. Shift Timings

Shift	Timing	Expected start hour (used for lateness check)
MORNING	6 AM – 2 PM	6
EVENING	2 PM – 10 PM	14
NIGHT	10 PM – 6 AM	22

6. System Design

6.1 Classes and Objects

Class / Enum	Role
Shift (enum)	Defines the three shift types: MORNING, EVENING, NIGHT.
Employee	Stores <code>id</code> , <code>name</code> and the assigned <code>shift</code> .
Attendance	One attendance record: employee ID, name, shift, check-in and check-out time. Calculates worked hours.
TimeUtil	Helper class that converts text such as "9:30 AM" to decimal hours (9.5) and back.
EmployeeManager	Business logic: holds all collections and implements every module.
EmployeeShiftGUI	The Swing window (extends <code>JFrame</code>); handles buttons, dialogs and output display.

6.2 Class Relationship

- `EmployeeShiftGUI` has an `EmployeeManager`, and calls its methods when a button is clicked.

- `EmployeeManager` **has many** `Employee` objects (`ArrayList`) and many `Attendance` records (`LinkedList`).
- `Attendance` copies the employee's ID, name and shift at the time of check-in.
- `Employee` and `Attendance` both use the `Shift` enum.
- `EmployeeShiftGUI` **extends** `JFrame` (inheritance).

6.3 How time is stored

Times are stored as decimal hours in 24-hour format. For example, 9:30 AM is stored as 9.5 and 5:45 PM as 17.75. The user types normal text ("9:30 AM"), `TimeUtil.parse()` converts it, and `TimeUtil.format()` converts it back for display. Worked hours are then a simple subtraction: `checkOut - checkIn`.

7. Java Concepts Used

Java Concept	Required Application	Where it is used in this project
Classes & Objects	<code>Employee</code> , <code>Shift</code> , <code>Attendance</code>	<code>Employee</code> , <code>Attendance</code> , <code>EmployeeManager</code> , <code>TimeUtil</code> classes
Constructors	Initialize shift/attendance objects	<code>Employee(int, String)</code> , <code>Attendance(Employee, double)</code> , <code>EmployeeManager()</code>
Enum	Shift type	<code>enum Shift { MORNING, EVENING, NIGHT }</code>
<code>StringBuilder</code>	Attendance report generation	<code>calculateHours</code> , <code>irregularities</code> , <code>searchByEmployee</code> , <code>searchByShift</code> , <code>generateReport</code> , <code>viewEmployees</code> , <code>viewAttendance</code>

Java Concept	Required Application	Where it is used in this project
ArrayList	Store employees	<code>ArrayList<Employee> employees</code>
LinkedList	Maintain attendance history	<code>LinkedList<Attendance> records</code>
HashMap	Shift details	<code>HashMap<Shift, String> shiftTimings</code> (shift → timing text)
TreeMap	Sorted report	<code>TreeMap<Integer, String> report</code> keeps the shift report sorted by employee ID
CRUD	Employee management	Create (<code>addEmployee</code>), Read (<code>viewEmployees</code>), Update (<code>updateEmployee</code>), Delete (<code>deleteEmployee</code>)
Searching	Search by employee/shift	<code>searchByEmployee</code> , <code>searchByShift</code> , <code>findEmployee</code>
Swing	Shift management GUI	<code>JFrame</code> , <code>JButton</code> , <code>JTextArea</code> , <code>JScrollPane</code> , <code>JOptionPane</code> , <code>GridLayout</code> , <code>BorderLayout</code>
Exception Handling	Invalid check-in/check-out data	<code>try/catch</code> for <code>NumberFormatException</code> in <code>getId</code> , <code>getTime</code> , <code>registerEmployee</code>
Validation	Shift timing, employee ID, hours	ID must be positive and unique; name cannot be empty; time must be valid H:MM AM/PM; shift must be assigned before check-in

8. Detailed Working of Each Module

8.1 Employee Registration (CRUD)

The administrator enters an employee ID and name. The system checks that the ID is a positive number, the name is not empty, and the ID is not already in use. If all checks pass, a new `Employee` is added to the `ArrayList`. Employees can also be viewed in a table, renamed (Update) or removed (Delete). Deleting looks up the employee by ID using `findEmployee()`.

8.2 Shift Scheduling

The administrator enters an employee ID and picks a shift from a drop-down list built from `Shift.values()`. The chosen shift is stored in the employee object. The **View Shifts** button prints the timing for each shift from the `HashMap`.

8.3 Attendance Recording

Check-in: the employee must exist and must already have a shift. The system also refuses a second check-in if the employee has an open record (check-out still pending). A new `Attendance` object is added to the `LinkedList` with `checkOut = -1`, which means "not checked out yet".

Check-out: the system finds the open record for that employee and stores the check-out time. If no open record exists, it shows "No active check-in found."

8.4 Worked Hours Calculation

For the entered employee ID, the system loops through all of that employee's records and prints worked hours for each. An incomplete record (no check-out) counts as 0 hours.

8.5 Irregularity Detection

Every record is checked against two rules:

1. **Late check-in:** check-in is more than 0.2 hours (about 12 minutes) after the shift's expected start time.
2. **Short shift:** the record is complete and worked hours are less than 8.

Each violation is listed with the employee's name.

8.6 Attendance Search

Records can be searched by employee ID or by shift type. Results are shown as an aligned table with ID, name, shift, in-time, out-time and hours.

8.7 Shift Report

For every registered employee, the system adds up total worked hours across all records and prints a summary of name, shift and total hours. The report is stored in a TreeMap so it always appears in ascending order of employee ID.

9. GUI Design

The window is divided into three areas using `BorderLayout`:

- **North:** a dark-blue heading with the application title.
- **Centre:** a scrollable, read-only output area with a monospaced font so table columns line up.
- **South:** a 5×3 grid of 14 buttons, one for each operation.

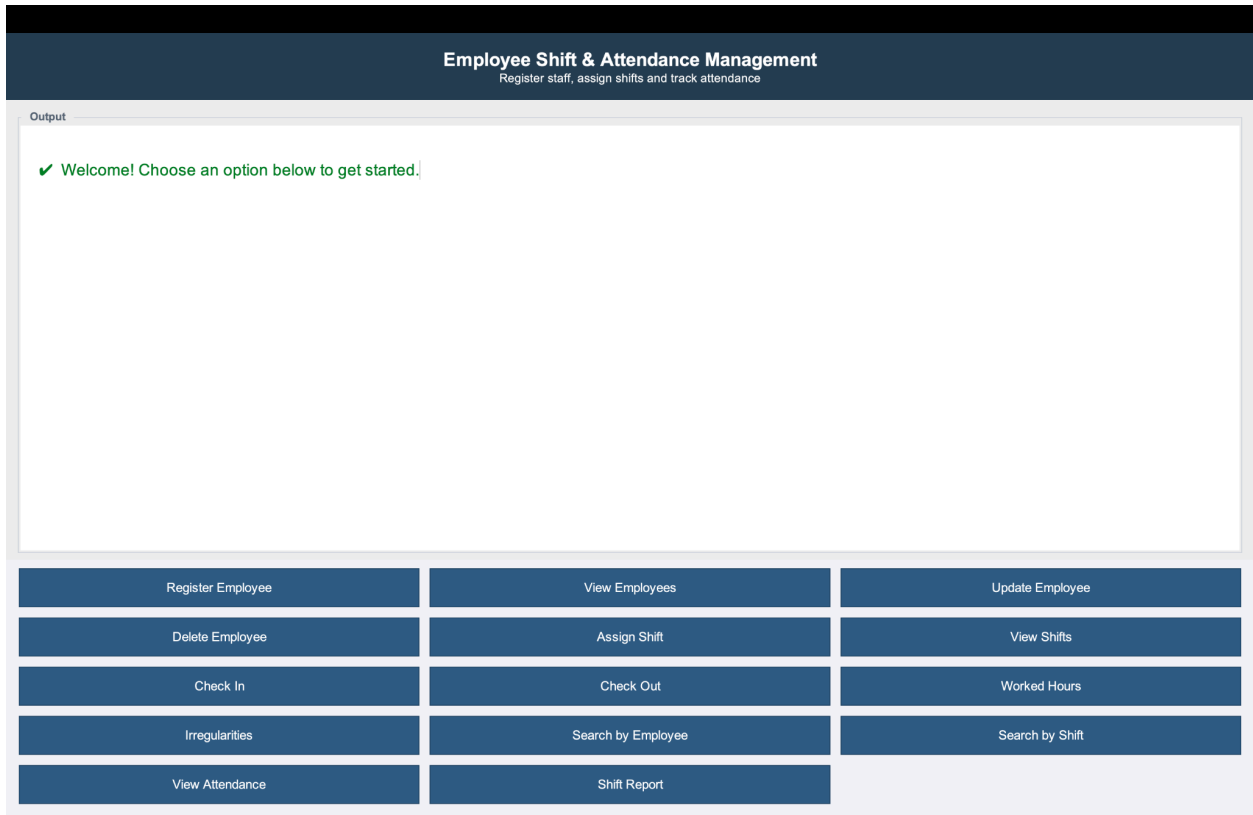
Status messages are coloured: green with a tick for success, red with a cross for errors, grey with an info icon for neutral messages. Input is collected through `JOptionPane` dialogs.

10. Exception Handling and Validation

Situation	How it is handled	Message shown
Non-numeric employee ID	<code>NumberFormatException</code> caught	"Please enter a numeric employee ID." / "Employee ID must be a number."
Invalid time format (e.g. "25:90 PM", "abc")	<code>TimeUtil.parse</code> throws, caught in <code>getTime</code>	"Please enter time like 9:30 AM or 5:45 PM."
Duplicate employee ID	Checked with <code>findEmployee</code>	"Employee ID already exists."
Empty name or ID ≤ 0	Validation in <code>addEmployee</code>	"Enter a valid ID and name."
Unknown employee ID	<code>findEmployee</code> returns null	"Employee not found."
Check-in without a shift	Checked in <code>checkIn</code>	"Assign a shift first."

Situation	How it is handled	Message shown
Double check-in	Open record detected	"Employee is already checked in."
Check-out without check-in	No open record	"No active check-in found."
Dialog cancelled	null returned and ignored	No action

11. Screenshots



Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

✓ Employee registered successfully.

?

Register Employee

Employee ID:

2

Employee Name:

Molick Jain

OK

Cancel

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ REGISTERED EMPLOYEES

ID	NAME	SHIFT
1	Manan Jain	MORNING
2	Molick Jain	Not assigned

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ REGISTERED EMPLOYEES

ID	NAME	SHIFT
1	Manan Jain	MORNING
2	Molick Jain	Not assigned

Input

Enter new name:
Karan Jain

OKCancel

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ REGISTERED EMPLOYEES

ID	NAME	SHIFT
1	Manan Jain	MORNING
2	Karan Jain	Not assigned

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

✓ Employee deleted successfully.

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ REGISTERED EMPLOYEES

ID	NAME	SHIFT
1	Manan Jain	MORNING

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

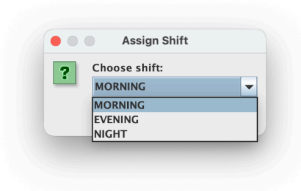
Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ REGISTERED EMPLOYEES

ID	NAME	SHIFT
1	Manan Jain	MORNING
2	Molick Jain	Not assigned
3	Naman Jain	Not assigned
4	Sanskriti Jain	Not assigned
5	Khushi	Not assigned



Register Employee	View Employees	Update Employee
Delete Employee	Assign Shift	View Shifts
Check In	Check Out	Worked Hours
Irregularities	Search by Employee	Search by Shift
View Attendance	Shift Report	



Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ SHIFT TIMINGS

MORNING

6 AM - 2 PM

EVENING

2 PM - 10 PM

NIGHT

10 PM - 6 AM

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ SHIFT TIMINGS

MORNING

6 AM - 2 PM

EVENING

2 PM - 10 PM

NIGHT

10 PM - 6 AM

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Input

?

Enter check-in time (e.g. 9:30 AM):

6:00 AM

⚙

Cancel

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

✓ Check-out recorded.

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ WORKED HOURS – Employee ID 1

Record 18.00 hours

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ IRREGULARITIES

* Molick Jain checked in late.
* Molick Jain worked less than 8 hours.

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ ATTENDANCE – Employee ID 1

ID	NAME	SHIFT	IN	OUT	HOURS
1	Manan Jain	MORNING	6:00 AM	2:00 PM	8.00

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ ATTENDANCE – EVENING Shift

ID	NAME	SHIFT	IN	OUT	HOURS
2	Molick Jain	EVENING	2:15 PM	6:00 PM	3.75

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

■ ATTENDANCE RECORDS

ID	NAME	SHIFT	IN	OUT	HOURS
1	Manan Jain	MORNING	6:00 AM	2:00 PM	8.00
2	Molick Jain	EVENING	2:15 PM	6:00 PM	3.75

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

Employee Shift & Attendance Management

Register staff, assign shifts and track attendance

Output

EMPLOYEE SHIFT REPORT

ID	NAME	SHIFT	HOURS
1	Manan Jain	MORNING	8.00
2	Molick Jain	EVENING	3.75
3	Naman Jain	NIGHT	0.00
4	Sanskriti Jain	EVENING	0.00
5	Khushi	NIGHT	0.00

Register Employee

View Employees

Update Employee

Delete Employee

Assign Shift

View Shifts

Check In

Check Out

Worked Hours

Irregularities

Search by Employee

Search by Shift

View Attendance

Shift Report

12. Sample Test Cases

No.	Action	Input	Expected result
1	Register employee	ID 101, Name "Rahul"	Employee registered successfully.
2	Register duplicate	ID 101 again	Employee ID already exists.
3	Register with text ID	ID "abc"	Employee ID must be a number.
4	Check-in without shift	ID 101	Assign a shift first.
5	Assign shift	ID 101, MORNING	Shift assigned successfully.
6	Check-in	ID 101, 6:00 AM	Check-in recorded.

No.	Action	Input	Expected result
7	Check-in again	ID 101, 7:00 AM	Employee is already checked in.
8	Check-out	ID 101, 2:00 PM	Check-out recorded.
9	Worked hours	ID 101	8.00 hours
10	Late check-in	ID 102 (Morning), 6:45 AM	Listed under irregularities as late.
11	Short shift	ID 103 (Morning), 6:00 AM – 11:00 AM	Listed as worked less than 8 hours.
12	Invalid time	"25:00 PM"	Please enter time like 9:30 AM or 5:45 PM.

13. Limitations and Future Enhancements

- **Data is not saved.** All records are held in memory and are lost when the program closes. File or database storage (JDBC/MySQL) can be added.
- **Night shift across midnight.** A night shift that starts at 10 PM and ends at 6 AM would currently give a negative worked-hours value, because check-out time is subtracted directly from check-in time on the same day. Adding 24 to the check-out time when it is smaller than the check-in time would fix this.
- **Sorting.** The case study mentions sorting attendance by hours worked and by shift type. This can be added using `Collections.sort()` with a `Comparator`, or a `TreeMap` keyed by worked hours.
- **Dates.** Attendance is recorded by time only, not by calendar date. Adding a date field would allow daily and monthly reports.
- **Login and roles.** An admin login can be added to restrict who may change records.

14. Conclusion

The Employee Shift & Attendance Management System successfully covers the main requirements of the case study. It uses an enum for shift categories, ArrayList and LinkedList for storing employees and attendance history, HashMap and TreeMap for shift details and sorted reports, and StringBuilder for generating

reports. Validation and exception handling keep invalid input from breaking the program, and the Swing interface makes the system easy to use. With the enhancements listed above, it could grow into a complete workforce management tool.